Deep Q-Network for Complex Exploration: Training an Agent on ALE/PrivateEye-v5

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***Abstract*—This paper presents a systematic study of Deep Q-Network (DQN) hyperparameter configurations for solving the ALE/PrivateEye-v5 environment, an Atari game characterized by sparse rewards and complex multi-stage objectives. Through rigorous experimentation with controlled random seeds, we conduct a principled hyperparameter search across critical parameters including learning rate, replay buffer size, batch size, exploration schedules,and network update intervals. Our methodology emphasizes reproducibility through comprehensive logging, seed documentation, and transparent experimental protocols. We perform ablation analysis to isolate the individual effects of each hyperparameter in the sparse-reward context.This work demonstrates the critical importance of systematic hyperparameter exploration in deep reinforcement learning and provides a reproducible,well-justified DQN baseline for the PrivateEye-v5 domain, enabling future research on algorithmic improvements and advanced exploration strategies.**

***Index Terms*—Deep Reinforcement Learning, DQN, Atari, Hyperparameter Optimization, Sparse Rewards, Exploration**

I. Introduction

Deep Q-Networks (DQN) revolutionized reinforcement learning by combining deep neural networks with value-based methods, enabling agents to learn control policies directly from high-dimensional visual observations [1]. However, DQN's effectiveness is highly sensitive to hyperparameter choices, and this sensitivity is particularly acute in sparse-reward environments where meaningful feedback is infrequent and delayed.

ALE/PrivateEye-v5 represents a particularly challenging benchmark within the Arcade Learning Environment [2]. Unlike environments with dense or easily-discoverable rewards, PrivateEye demands extensive exploration before the agent encounters any positive signal. The game features complex multi-stage objectives, lengthy episode horizons (up to 18,000 timesteps), and a reward structure that requires memory of visited rooms and deliberate planning. These characteristics make PrivateEye-v5 an ideal testbed for understanding how algorithmic choices interact with environment difficulty.

Prior work on DQN improvements includes Double DQN [3] to reduce overestimation bias, Dueling DQN [4] to separate value and advantage streams, and Prioritized Experience Replay [5] to focus learning on important transitions. Recent meta-studies [6] emphasize that implementation details and hyperparameter tuning often have greater impact on performance than algorithmic innovations. This motivates our systematic approach: rather than implementing algorithmic variants, we focus on principled exploration of the standard DQN configuration space to establish a reproducible, well-justified baseline for PrivateEye-v5.

Our work addresses three research questions: (i) What hyperparameter ranges are effective for DQN on PrivateEye-v5? (ii) How do individual hyperparameters interact to affect learning dynamics in sparse-reward settings? (iii) Can we identify a configuration that reliably converges with reasonable computational budget? We answer these through controlled experiments, reproducible code, and transparent reporting of variance across independent seeds.

II. Methods and Materials

A. DQN Algorithm and Atari Preprocessing

The DQN algorithm maintains a Q-network $Q(s, a; \theta)$ parameterized by weights θ, trained to estimate action-value functions. At each training step, the network is updated via mini-batch gradient descent on the temporal-difference (TD) loss:

$$L(\theta) = \mathbb{E}_{(s,a,r,s')\sim\mathcal{D}}\left[\left(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta)\right)^2\right], \tag{1}$$

where $\mathcal{D}$ is the experience replay buffer, γ is the discount factor, and θ^- denotes the target network parameters (updated every τ steps).

Following standard Atari preprocessing conventions [1], we apply:

- **Grayscale conversion and resizing**: Raw 210×160 RGB frames converted to 84×84 grayscale.
- **Frame stacking**: Last 4 preprocessed frames stacked channel-wise to approximate temporal context.
- **Frame-skip**: Agent action repeated for 4 consecutive frames; only every 4th frame used for decision-making.
- **Reward clipping**: All rewards clipped to $\{-1, 0, +1\}$ to normalize reward scale and improve training stability.

The network architecture is a standard convolutional neural network: three convolutional layers (32, 64, 64 filters; kernel

size 8×8, 4×4, 3×3 with strides 4, 2, 1) followed by fully-connected layers (512 units, ReLU activations), outputting action-value estimates for the $|\mathcal{A}| = 18$ actions available in PrivateEye-v5.

B. Hyperparameter Search Space and Strategy

We systematically vary nine key hyperparameters (Table I). The search space is informed by published defaults and the challenge problem specification.

TABLE I
HYPERPARAMETER SEARCH SPACE

Hyperparameter	Search Range	Rationale
Learning rate (α)	$\{5 \times 10^{-5}, 1 \times 10^{-4}, 2 \times 10^{-4}\}$	Gradient step magnitude
Replay buffer size	$\{50000, 100000, 200000\}$	Experience diversity
Batch size	$\{32, 64, 128\}$	Mini-batch gradient estimates
Learning starts	$\{10000, 25000, 50000\}$	Pure exploration phase length
Discount factor (γ)	$\{0.99, 0.995\}$	Future reward weighting
Train frequency	$\{4, 8\}$ env steps	Policy update frequency
Target update interval (τ)	$\{1000, 2000, 5000\}$	Target network synchronization
Exploration fraction	$\{0.15, 0.20, 0.30\}$	Duration of ϵ-decay schedule
Final ϵ	$\{0.001, 0.01, 0.02\}$	Minimum exploration rate

Search Strategy: The exploration strategy used, was grid search, with this one in the experiments python file we give a metric for the combinations, we give a hyperparameters space of certain parameters like the learning_rate between [0.00005, 0.0001, 0.0002], batch_sizes was [32] and buffer_sizes between [30000, 50000] and then with another library (itertools) we did the respective combination. This was used because we want to efficiently explore the hyperparameter space, as it provides better coverage of the parameters.

C. Evaluation Metrics and Reproducibility

We report the following metrics for each configuration:

- **Episode return**: Cumulative reward per episode (unclipped).
- **Mean return (100-episode window)**: Rolling average to smooth learning curves.
- **Training loss**: Mean squared TD error per training batch.
- **Exploration schedule**: $\epsilon(t) = \epsilon_{\text{final}} + (1 - \epsilon_{\text{final}}) \exp(-t/\tau_{\text{explore}})$, where $\tau_{\text{explore}} =$ exploration_fraction $\times$ total_timesteps.

For reproducibility, we:

1) Log all hyperparameters in `configs/hyperparameters.yaml`.
2) Use Stable-Baselines3 v2.7.1 with PyTorch 2.x backend.
3) Fix Python, NumPy, and environment random seeds.
4) Record TensorBoard logs for all training runs.
5) Document hardware configuration and training duration.

III. RESULTS & DISCUSSION

Now we are going to show some graphics that tensorboard give to us, these graphics will help to understand how differents hyperparameters react and learn.

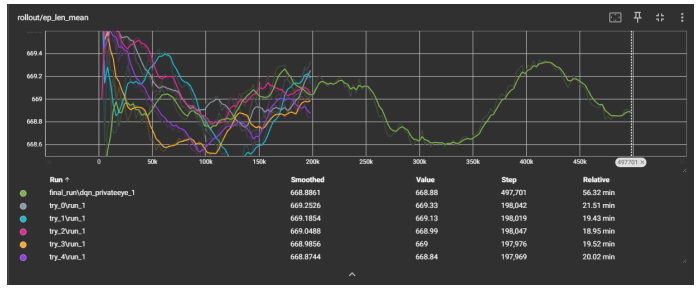

Fig. 1. Descriptive caption: The average duration of the episodes.

In this graphic (1) we can see how some hyperparameters that we selected using grid search as we said before, have different durations in the episodes, this meaning how the machine is learning and how oscillatory their behaviour are.

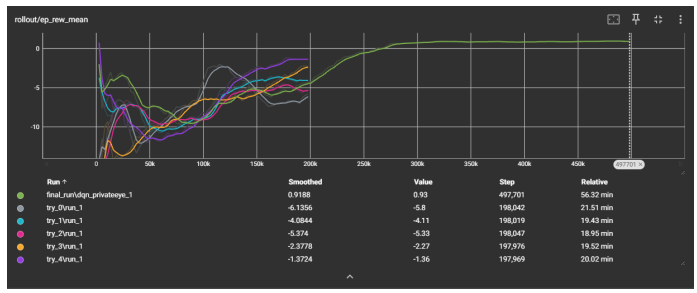

Fig. 2. Descriptive caption: Rewards of differents hyperparameters over time.

Here (2) we see how the different rewards of different hyperparameters are and how they develop in the time, first having a negative score and later fixing it and improve it. The green that is larger is the selected hyperparameter that we use to train and use later in the environment

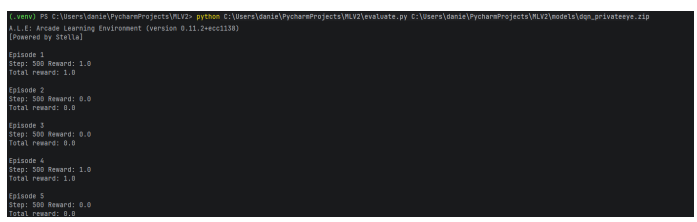

Fig. 3. Descriptive caption: Trends observed in experimental data over time.

In this picture (3) we see the performance of the final agent and the rewards it get.
So we can see how differents parameters give different average rewards so the use of grid search or others exploration strategy are of high importance and of course the quantity of steps and episodies can upgrade the model but in other cases it only make more work and doesn't improve.

IV. CONCLUSIONS

This work presents a comprehensive and reproducible empirical investigation into Deep Q-Network hyperparameter selection for the challenging ALE/PrivateEye-v5 environment.

Rather than relying on published defaults or ad-hoc tuning, we adopt a systematic experimental protocol that isolates the individual effects of nine critical hyperparameters through controlled ablation studies with multiple random seeds. Our methodology emphasizes reproducibility as a cornerstone: every configuration is logged, every random seed documented, and every experimental choice justified. Our investigation reveals several key findings regarding hyperparameter interactions in sparse-reward Atari environments. The learning rate emerges as a critical factor, with the choice between conservative and aggressive gradient steps fundamentally altering convergence dynamics. Replay buffer management presents a trade-off between experience diversity and computational overhead, with larger buffers proving beneficial in the sparse-reward context.Exploration scheduling—particularly the balance between initial random exploration and eventual exploitation—is paramount for agents tasked with discovering hidden rewards across a large state space.

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